

Support Tensor Theory for Additive Manufacturing Orientation Optimization

InHwan Sul et al.

Abstract

This paper proposes a novel tensor-based framework for rapid build orientation estimation in additive manufacturing. The proposed Support Tensor converts the nonlinear support-generation problem into an eigenvalue problem on a symmetric positive semidefinite matrix. The method enables rapid identification of candidate build directions directly from STL geometry without exhaustive directional search.

Keywords: Additive Manufacturing, Build Orientation, Support Structure, Eigenvalue Analysis, PCA, Tensor Methods

1 Introduction

Build orientation strongly influences support volume, build time, material consumption, surface quality, and post-processing cost in additive manufacturing.

Current approaches typically rely on:

1. exhaustive directional sampling,
2. genetic algorithms,
3. topology-aware search,
4. commercial slicer simulation.

These approaches are computationally expensive.

This study proposes a new mathematical descriptor called the *Support Tensor*. The key idea is to represent support tendency through a second-order tensor derived from surface-normal distributions.

2 Related Work

PCA has long been used for geometric analysis.

For a point cloud,

$$C = \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \quad (1)$$

and eigenvectors of C define principal directions.

However, support generation depends primarily on surface orientation rather than point distribution.

3 Support Functional

Let

- A_i : triangle area
- $\mathbf{m}_i$: unit outward normal
- $\mathbf{n}$: build direction

The support tendency is approximated by

$$S(\mathbf{n}) = \sum_i A_i H(-\mathbf{m}_i \cdot \mathbf{n})(-\mathbf{m}_i \cdot \mathbf{n}). \quad (2)$$

This function is nonlinear and difficult to optimize directly.

4 Quadratic Approximation

Consider

$$\tilde{S}(\mathbf{n}) = \sum_i A_i (\mathbf{m}_i \cdot \mathbf{n})^2. \quad (3)$$

Since

$$(\mathbf{m}_i \cdot \mathbf{n})^2 = \mathbf{n}^T (\mathbf{m}_i \mathbf{m}_i^T) \mathbf{n}, \quad (4)$$

we obtain

$$\tilde{S}(\mathbf{n}) = \mathbf{n}^T M \mathbf{n}. \quad (5)$$

Definition 1 (Support Tensor).

$$M = \sum_i A_i \mathbf{m}_i \mathbf{m}_i^T. \quad (6)$$

5 Theoretical Properties

Proposition 1. *The Support Tensor is symmetric.*

Proof

Each term $\mathbf{m}_i \mathbf{m}_i^T$ is symmetric.

Therefore

$$M^T = M. \quad (7)$$

□

Proposition 2. *The Support Tensor is positive semidefinite.*

Proof

For any vector $\mathbf{x}$,

$$\mathbf{x}^T M \mathbf{x} = \sum_i A_i (\mathbf{x}^T \mathbf{m}_i)^2 \geq 0. \quad (8)$$

□

Theorem 1. *The minimum of*

$$\mathbf{n}^T M \mathbf{n} \quad (9)$$

under

$$\|\mathbf{n}\| = 1 \quad (10)$$

is the smallest eigenvalue of M .

Proof

Directly follows from the Rayleigh Quotient theorem.

□

6 Weighted Support Tensor

A generalized model is

$$M_w = \sum_i w_i A_i \mathbf{m}_i \mathbf{m}_i^T. \quad (11)$$

where

$$w_i = f(q_i, d_i, r_i). \quad (12)$$

Possible weighting factors:

- cavity influence
- overhang sensitivity
- support accessibility
- local manufacturing penalty

7 Cavity-Aware Extension

Let

$$q_i \in [0, 1] \quad (13)$$

be a cavity score.

Then

$$w_i = 1 + \alpha q_i. \quad (14)$$

This gives

$$M_{cavity} = \sum_i (1 + \alpha q_i) A_i \mathbf{m}_i \mathbf{m}_i^T. \quad (15)$$

The eigenvectors become sensitive to internal voids and trapped regions.

8 Algorithm

Algorithm 1 Support Tensor Orientation Estimation

```

for each triangle do
  Compute area
  Compute unit normal
  Update tensor
end for
Eigen decomposition
Select minimum-eigenvalue eigenvector
Validate with support simulation

```

9 Experimental Design

Benchmark models:

1. Stanford Bunny
2. Mechanical Bracket
3. Turbine Blade
4. Thermal Manikin Components
5. Apparel-related 3D Models

Metrics:

$$E_{angle} = \cos^{-1}(|\mathbf{n}_{pred} \cdot \mathbf{n}_{best}|) \quad (16)$$

and

$$E_{support} = \frac{S(\mathbf{n}_{pred}) - S(\mathbf{n}_{best})}{S(\mathbf{n}_{best})}. \quad (17)$$

10 Discussion

The proposed approach has several advantages:

1. Extremely low computational cost.
2. Closed-form eigenvalue solution.

3. Easy STL implementation.
4. Extendable to cavity-aware optimization.

Potential limitations include:

1. inability to fully model support visibility,
2. slicer-specific rules,
3. self-shadowing effects.

11 Future Research

Future work includes:

- support visibility tensors,
- graph-based cavity descriptors,
- thermal distortion weighting,
- multi-objective optimization,
- topology-aware support tensors.

12 Conclusion

Support Tensor Theory transforms orientation optimization into a tractable eigenvalue problem. The framework provides a new connection between tensor analysis, PCA, and additive manufacturing orientation selection.

References

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